

Microvolt T-Wave Alternans (MTWA)

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Tracking Information

Publication Number

100-3

Manual Section Number

20.30

Manual Section Title

Microvolt T-Wave Alternans (MTWA)

Version Number

3

Effective Date of this Version

01/13/2015

Implementation Date

06/23/2015

Description Information

Benefit Category

Diagnostic Tests (other)

Please Note: This may not be an exhaustive list of all applicable Medicare benefit categories for this item or service.

Item/Service Description

A. General

Microvolt T-wave Alternans (MTWA) testing is a non-invasive diagnostic test that detects minute electrical activity in a portion of the electrocardiogram (ECG) known as the T-wave. MTWA testing has a role in the stratification of patients who may be at risk for sudden cardiac death (SCD) from ventricular arrhythmias.

Within patient groups that may be considered candidates for implantable cardioverter defibrillator (ICD) therapy, a negative MTWA test may be useful in identifying low-risk patients who are unlikely to benefit from, and who may experience worse outcomes from, ICD placement.

Spectral analysis (SA) is a sensitive mathematical method of measuring and comparing time and the ECG signals. It requires specialized propriety electrodes to calculate minute T-wave voltage changes. Software then analyzes these microvolt changes and produces a report to be interpreted by a physician. The Modified Moving Average (MMA) method uses a temporal domain in which T-wave alternans are assessed as a continuous variable along the complete ECG. The MMA method of MTWA testing is performed using standard ambulatory ECG equipment.

Indications and Limitations of Coverage

B. Nationally Covered Indications

Effective for dates of service on and after March 21, 2006, MTWA diagnostic testing is covered for the evaluation of patients at risk for SCD, only when the SA method is used.

C. Nationally Non-Covered Indications

N/A

D. Other

Effective for dates of service on and after January 21, 2015, Medicare Administrative Contractors (MACs) acting within their respective jurisdictions may determine coverage of MTWA diagnostic testing for the evaluation of patients at risk for SCD using all other methods.

(This NCD last reviewed January 2015.)

Claims Processing Instructions

[TN 3265 \(Medicare Claims Processing\)](#) 

Transmittal Information

Transmittal Number

182

Coverage Transmittal Link

<http://www.cms.gov/Regulations-and-Guidance/Guidance/transmittals/downloads/R182NCD.pdf> 

Revision History

01/2024 - Transmittal 12318 issued October 19, 2023, is being rescinded and replaced by Transmittal 12441, dated January 3, 2024, to add FISS to BR 13390.11 and to add clarifying verbiage to business requirement 13390.12. All other information remains the same. ([TN 12441](#) ) (CR13390)

10/2023 - The purpose of this Change Request (CR) is to provide a quarterly maintenance update of ICD-10 coding conversions and other coding updates specific to National Coverage Determinations (NCDs). No policy is being changed as a result of these updates. ([TN 12318](#) ) (CR13390)

08/2022 - Transmittal 11545 dated August 5, 2022, is being rescinded and replaced by Transmittal 11584, dated, August 31, 2022. ([TN 11584](#)) (CR12822)

08/2022 - The purpose of this Change Request (CR) is to provide a maintenance update of ICD-10 conversions and other coding updates to specific NCDs. ([TN 11545](#)) (CR12822)

01/2018 - This Change Request (CR) constitutes a maintenance update of International Code of Diseases, Tenth Revision (ICD-10) conversions and other coding updates specific to National Coverage Determinations (NCDs). These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received.

Previous NCD coding changes appear in ICD-10 quarterly updates that can be found at: <https://www.cms.gov/Medicare/Coverage/CoverageGenInfo/ICD10.html>, along with other CRs implementing new policy NCDs. Edits to ICD-10 and other coding updates specific to NCDs will be included in subsequent quarterly releases and individual CRs as appropriate. No policy-related changes are included with the ICD-10 quarterly updates. Any policy-related changes to NCDs continue to be implemented via the current, long-standing NCD process. ([TN 2005](#)) (CR10318)

11/2017 - This Change Request (CR) constitutes a maintenance update of International Code of Diseases, Tenth Revision (ICD-10) conversions and other coding updates specific to National Coverage Determinations (NCDs). These NCD coding changes are the result of newly available codes, coding revisions to NCDs released separately, or coding feedback received. ([TN 1975](#)) (CR10318)

05/2015 - Effective January 13, 2015, CMS decided that no national coverage determination (NCD) is appropriate at this time for for microvolt t-wave alternans (MTWA) testing using the modified moving average (MMA) method for the evaluation of patients at risk for sudden cardiac death (SCD). Implementation date: 06/23/2015 Effective date: 01/13/2015 ([TN 182](#)) (CR9162)

03/2013 - CMS translated the information for this policy from ICD-9-CM/PCS to ICD-10-CM/PCS according to HIPAA standard medical data code set requirements and updated any necessary and related coding infrastructure. These updates do not expand, restrict, or alter existing coverage policy. Implementation date: 10/07/2013 Effective date: 10/1/2015. ([TN 1199](#)) ([TN 1199](#)) (CR 8197)

07/2008 - Effective Date: 05/12/2008. Implementation Date: 08/25/2008. ([TN 88](#)) (CR6154)

03/2006 - Effective Date: 03/21/2006. Implementation Date: 04/03/2006. ([TN 49](#)) (CR4351)

National Coverage Analyses (NCAs)

This NCD has been or is currently being reviewed under the National Coverage Determination process. The following are existing associations with NCAs, from the National Coverage Analyses database.

- [Original Consideration for Microvolt T-wave Alternans \(CAG-00293N\)](#) 
- [First reconsideration for Microvolt T-wave Alternans \(CAG-00293R\)](#) 
- [Second reconsideration for Microvolt T-wave Alternans \(CAG-00293R2\)](#) 

Additional Information

Other Versions

Title	Version	Effective Between	
Microvolt T-Wave Alternans (MTWA)	3	01/13/2015 - N/A	You are here
Microvolt T-Wave Alternans (MTWA)	2	05/12/2008 - 01/13/2015	View
Microvolt T-Wave Alternans (MTWA)	1	03/21/2006 - 05/12/2008	View